Transforming the Canonical Lagrangian of a Relativistic Particle for a Smooth Massless Limit: The Einbein Formulation in $1 + 1$ Dimensions

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Technical Tutorial Notes for Mathematical Physics Students

(General Relativity, Lagrangian Mechanics, and the Massless Limit)

Abstract

This tutorial derives, step by step and without omitting intermediate algebraic manipulations, the reformulation of the canonical square-root action for a relativistic point particle in a generic $(1 + 1)$-dimensional Lorentzian metric that permits a smooth limit $m \to 0$. The key device is the introduction of an auxiliary world-line scalar density (the *einbein*). All Euler–Lagrange equations are computed explicitly, the algebraic solution for the einbein is substituted back into the action, and the recovery of the original square-root form is verified by direct calculation. The massless limit is then taken and shown to enforce the null condition while yielding the correct null-geodesic equation. The presentation is self-contained for a reader already familiar with the principles of general relativity, the Lagrangian formalism, and the Euler–Lagrange equations. References to standard textbooks and lecture notes are provided with author names, titles, publication years and, where available, stable URLs.

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1 Introduction and Motivation

The motion of a free relativistic point particle of rest mass m in a curved spacetime is governed by the principle of stationary proper time. In units where the speed of light is kept explicit and denoted c_0, the corresponding action functional is proportional to the proper length of the world-line:

$$S[x] = -\,mc_0 \int \sqrt{-g_{\mu\nu}(x)\,\dot{x}^\mu \dot{x}^\nu}\,\mathrm{d}\tau. \tag{1}$$

(The overall minus sign is conventional so that the action is negative for a particle at rest.) This expression is reparametrization-invariant and yields the geodesic equation upon variation. However, two practical difficulties arise:

(i) The presence of the square root makes the Lagrangian non-quadratic in the velocities, complicating both canonical analysis and path-integral quantization.

(ii) When the rest mass vanishes, $m \to 0$, the integrand vanishes identically on every null trajectory. The variational principle then becomes singular or trivial, and the correct massless dynamics cannot be recovered by a direct limit.

A standard remedy, widely used in the string-theory literature as a warm-up to the Polyakov action,[1] is to introduce an auxiliary field $e(\tau) > 0$ (the *einbein*) that converts the action into a quadratic expression. The resulting formulation remains equivalent to (1) for $m > 0$ and admits a perfectly regular massless limit.

The purpose of these notes is to carry out every algebraic step of this reformulation in the simplified setting of $1 + 1$ spacetime dimensions. No intermediate calculation is declared "obvious"; every partial derivative required by the Euler–Lagrange equations is written explicitly.

[1]The *Polyakov action* is the two-dimensional world-sheet analogue of the einbein formulation presented in these notes. It replaces the square-root Nambu–Goto action for a relativistic string by a quadratic action that involves an auxiliary world-sheet metric (the two-dimensional counterpart of the einbein). The formulation was introduced by A. M. Polyakov, "Quantum geometry of bosonic strings", *Phys. Lett.* B **103** (1981) 207–210, and is the starting point for the modern path-integral quantization of bosonic string theory. See also J. Polchinski, *String Theory*, Vol. 1 (Cambridge University Press, 1998), Chapter 3.

2 Notation and Geometric Setup in $1+1$ Dimensions

We work on a two-dimensional Lorentzian manifold $\mathcal{M}$ equipped with a metric tensor $g_{\mu\nu}$ of signature $(-,+)$. Local coordinates are denoted $x^\mu = (x^0, x^1)$. A world-line is a smooth map

$$x^\mu : \; I \subset \mathbb{R} \;\longrightarrow\; \mathcal{M}, \qquad \tau \mapsto x^\mu(\tau), \tag{2}$$

where τ is an arbitrary parameter (not necessarily proper time). Derivatives with respect to τ are written with a dot:

$$\dot{x}^\mu(\tau) \;:=\; \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}. \tag{3}$$

The induced interval on the world-line is

$$\mathrm{d}s^2 \;=\; g_{\mu\nu}(x(\tau))\,\mathrm{d}x^\mu\,\mathrm{d}x^\nu \;=\; g_{\mu\nu}(x)\,\dot{x}^\mu\dot{x}^\nu\,\mathrm{d}\tau^2. \tag{4}$$

We adopt the convention that a future-directed timelike vector v^μ satisfies

$$g_{\mu\nu}v^\mu v^\nu \;<\; 0. \tag{5}$$

Consequently the quantity

$$\sigma \;:=\; g_{\mu\nu}(x)\,\dot{x}^\mu\dot{x}^\nu \tag{6}$$

is negative for a timelike world-line. The determinant of the metric is negative,

$$\det(g_{\mu\nu}) \;=\; g_{00}g_{11} - g_{01}^2 \;<\; 0. \tag{7}$$

Throughout we keep the speed of light c_0 explicit. Rest mass is denoted $m > 0$.

3 The Canonical Square-Root Action

The action that extremizes proper time is

$$S_{\text{sqrt}}[x] \;=\; -\,mc_0 \int_{\tau_i}^{\tau_f} \sqrt{-\,g_{\mu\nu}(x)\,\dot{x}^\mu\dot{x}^\nu}\,\mathrm{d}\tau \;=\; -\,mc_0 \int \sqrt{-\sigma}\,\mathrm{d}\tau. \tag{8}$$

The associated Lagrangian density (with respect to the parameter τ) is therefore

$$L_{\text{sqrt}} \;=\; -\,mc_0\sqrt{-g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu} \;=\; -\,mc_0\sqrt{-\sigma}. \tag{9}$$

Because L_{sqrt} is homogeneous of degree one in the velocities, the action is invariant under arbitrary reparametrizations $\tau \to f(\tau)$ with $\dot{f} > 0$. The Euler–Lagrange equations obtained from L_{sqrt} are equivalent to the geodesic equation

$$\frac{\mathrm{D}\dot{x}^\mu}{\mathrm{d}\tau} \;=\; 0 \tag{10}$$

when the parameter is chosen to be an affine parameter (or, more generally, proportional to proper time).

4 Difficulties with the Massless Limit

If one attempts to set $m = 0$ directly in (8), the action vanishes for every trajectory that satisfies the null condition $\sigma = 0$. The variational principle then ceases to select a preferred class of curves: every null curve yields the same (zero) value of the action. Moreover, the square-root Lagrangian becomes non-differentiable on the light cone. Consequently a different formulation is required if one wishes a single action principle that covers both the massive and the massless cases continuously.

5 The Einbein Reformulation

Introduce a real auxiliary field $e(\tau) > 0$, called the *einbein*. Geometrically, e may be viewed as defining an intrinsic metric on the one-dimensional world-line via

$$\mathrm{d}s^2_{\text{worldline}} \;=\; e(\tau)^2\,\mathrm{d}\tau^2. \tag{11}$$

Consider the new action functional that depends on both the embedding $x^\mu(\tau)$ and the einbein $e(\tau)$:

$$S[x,e] \;=\; \int_{\tau_i}^{\tau_f} \left(\frac{1}{2e}\, g_{\mu\nu}(x)\,\dot{x}^\mu \dot{x}^\nu - \frac{e}{2}(mc_0)^2 \right) \mathrm{d}\tau. \tag{12}$$

The corresponding Lagrangian is

$$L(x,\dot{x},e) \;=\; \frac{1}{2e}\,\sigma - \frac{e}{2}(mc_0)^2, \tag{13}$$

where $\sigma = g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu$ as before. Note that L is quadratic in the velocities and contains no derivative of e; the einbein is therefore an auxiliary (non-dynamical) field.

6 Euler–Lagrange Equation for the Einbein

The general Euler–Lagrange equation associated with any variable $\phi(\tau)$ is

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\left(\frac{\partial L}{\partial \dot{\phi}}\right) - \frac{\partial L}{\partial \phi} \;=\; 0. \tag{14}$$

We now specialise this equation to the einbein by a sequence of purely logical steps.

Step 1. Start from the general Euler–Lagrange equation (14).

Step 2. Choose the variable $\phi = e(\tau)$. The equation becomes

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\left(\frac{\partial L}{\partial \dot{e}}\right) - \frac{\partial L}{\partial e} \;=\; 0. \tag{15}$$

Step 3. Inspection of the explicit Lagrangian

$$L = \frac{1}{2e}\, g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu - \frac{e}{2}(mc_0)^2 \tag{16}$$

shows that the derivative $\dot{e}$ never appears. Consequently the partial derivative with respect to that velocity vanishes identically:

$$\frac{\partial L}{\partial \dot{e}} \;\equiv\; 0. \tag{17}$$

Step 4. The ordinary derivative of the zero function is zero:

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\left(\frac{\partial L}{\partial \dot{e}}\right) = \frac{\mathrm{d}}{\mathrm{d}\tau}(0) = 0. \tag{18}$$

Step 5. Substituting the result of Step 4 into the specialised equation of Step 2 leaves only

$$-\frac{\partial L}{\partial e} \;=\; 0. \tag{19}$$

Step 6. Multiplying both sides by -1 (or simply rearranging the signs) yields the equivalent statement

$$\frac{\partial L}{\partial e} \;=\; 0. \tag{20}$$

Thus equation (20) is nothing but the general Euler–Lagrange equation after the observation that $\partial L/\partial \dot{e}$ is identically zero has been used. No further dynamical assumption is required; the reduction is purely formal. (This is the hallmark of an *auxiliary* or *non-dynamical* field: its equation of motion contains no derivatives of the field itself and can be solved algebraically.)

We now compute the partial derivative $\partial L/\partial e$ term by term. The Lagrangian is the sum of two pieces:

$$L \;=\; \underbrace{\frac{1}{2e}\,\sigma}_{L_1} + \underbrace{\Big(-\frac{e}{2}(mc_0)^2\Big)}_{L_2}. \tag{21}$$

Differentiating each piece separately with respect to e (treating the velocity combination σ as independent of e) yields

$$\frac{\partial L_1}{\partial e} = \frac{\partial}{\partial e}\Big(\frac{1}{2e}\sigma\Big) \;=\; \sigma \cdot \frac{\partial}{\partial e}\Big(\frac{1}{2e}\Big) \;=\; \sigma \cdot \Big(-\frac{1}{2e^2}\Big) \;=\; -\frac{1}{2e^2}\sigma, \tag{22}$$

$$\frac{\partial L_2}{\partial e} = \frac{\partial}{\partial e}\Big(-\frac{e}{2}(mc_0)^2\Big) \;=\; -\frac{1}{2}(mc_0)^2. \tag{23}$$

Because differentiation is a linear operation,

$$\frac{\partial L}{\partial e} \;=\; \frac{\partial L_1}{\partial e} + \frac{\partial L_2}{\partial e} \;=\; -\frac{1}{2e^2}\sigma - \frac{1}{2}(mc_0)^2. \tag{24}$$

The Euler–Lagrange condition $\partial L/\partial e = 0$ (already established in equation (20)) therefore requires

$$-\frac{1}{2e^2}\sigma - \frac{1}{2}(mc_0)^2 \;=\; 0. \tag{25}$$

This is the content of equation (19).

To solve the algebraic equation we multiply both sides by the nowhere-vanishing factor $-2e^2$:

$$(-2e^2)\left(-\frac{1}{2e^2}\sigma\right) + (-2e^2)\left(-\frac{1}{2}(mc_0)^2\right) \;=\; 0$$

$$\sigma + e^2(mc_0)^2 \;=\; 0. \tag{26}$$

Equivalently,

$$g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu \;=\; -e^2(mc_0)^2. \tag{27}$$

Solving for the positive root $e > 0$ gives

$$e(\tau) \;=\; \frac{1}{mc_0}\sqrt{-g_{\mu\nu}(x)\,\dot{x}^\mu\dot{x}^\nu} \;=\; \frac{\sqrt{-\sigma}}{mc_0}. \tag{28}$$

(The negative root would correspond to a time-reversed parametrization and is discarded by the orientation convention $e > 0$.)

7 Substitution and Recovery of the Square-Root Action

We now substitute the algebraic solution (28) back into the Lagrangian (13) and verify that the original square-root Lagrangian is recovered exactly.

From the constraint (27) we have

$$\sigma \;=\; -e^2(mc_0)^2. \tag{29}$$

Insert this expression into the first term of L:

$$\frac{1}{2e}\,\sigma = \frac{1}{2e}\big(-e^2(mc_0)^2\big) \;=\; -\frac{e}{2}(mc_0)^2. \tag{30}$$

Therefore the whole Lagrangian becomes

$$L = -\frac{e}{2}(mc_0)^2 - \frac{e}{2}(mc_0)^2 \;=\; -e\,(mc_0)^2. \tag{31}$$

Now replace e by its explicit solution (28):

$$L = -\Big(\frac{\sqrt{-\sigma}}{mc_0}\Big)(mc_0)^2 \;=\; -\,mc_0\sqrt{-\sigma} \;=\; -\,mc_0\sqrt{-g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu}. \tag{32}$$

This is precisely L_{sqrt} of equation (9). Consequently the two action principles are classically equivalent for $m > 0$: every critical point of $S[x, e]$ that satisfies the einbein equation of motion projects to a critical point of $S_{\text{sqrt}}[x]$, and vice versa.

8 Euler–Lagrange Equations for the Coordinates (Massive Case)

We next derive the equations of motion for the embedding coordinates $x^\lambda(\tau)$ in the massive case $m > 0$. The Lagrangian

$$L(x, \dot{x}, e) \;=\; \frac{1}{2e}\,g_{\mu\nu}(x)\,\dot{x}^\mu\dot{x}^\nu - \frac{e}{2}(mc_0)^2 \tag{33}$$

depends on the coordinates both through the metric coefficients $g_{\mu\nu}(x)$ and through the velocities $\dot{x}^\mu$. The Euler–Lagrange equation for each component x^λ ($\lambda = 0, 1$) is

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\Big(\frac{\partial L}{\partial \dot{x}^\lambda}\Big) \;=\; \frac{\partial L}{\partial x^\lambda}. \tag{34}$$

Throughout this section the einbein $e(\tau)$ is treated as an independent field (its own equation of motion will be imposed afterwards).

8.1 Partial derivative with respect to the velocity

Differentiate L with respect to $\dot{x}^\lambda$. Only the first term contributes:

$$\begin{aligned}\frac{\partial L}{\partial \dot{x}^\lambda} &= \frac{\partial}{\partial \dot{x}^\lambda}\left(\frac{1}{2e}\,g_{\mu\nu}(x)\,\dot{x}^\mu\dot{x}^\nu\right)\\ &= \frac{1}{2e}\,g_{\mu\nu}(x)\Big(\delta^\mu{}_\lambda\dot{x}^\nu + \dot{x}^\mu\delta^\nu{}_\lambda\Big)\\ &= \frac{1}{2e}\big(g_{\lambda\nu}\dot{x}^\nu + g_{\mu\lambda}\dot{x}^\mu\big)\\ &= \frac{1}{e}\,g_{\lambda\nu}(x)\,\dot{x}^\nu,\end{aligned} \tag{35}$$

where the last equality follows from the symmetry $g_{\mu\lambda} = g_{\lambda\mu}$. Consequently the left-hand side of the Euler–Lagrange equation is the total τ-derivative

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\Big(\frac{\partial L}{\partial \dot{x}^\lambda}\Big) \;=\; \frac{\mathrm{d}}{\mathrm{d}\tau}\Big(\frac{1}{e}\,g_{\lambda\nu}(x)\,\dot{x}^\nu\Big). \tag{36}$$

8.2 Partial derivative with respect to the coordinate

At fixed velocities the only dependence of L on the coordinate x^λ comes from the metric tensor:

$$\frac{\partial L}{\partial x^\lambda} \;=\; \frac{1}{2e}\,\big(\partial_\lambda g_{\mu\nu}(x)\big)\dot{x}^\mu\dot{x}^\nu. \tag{37}$$

(The second term of L does not depend on x.)

8.3 The Euler–Lagrange equation before imposing the einbein constraint

Equating (36) and (37) yields the explicit Euler–Lagrange equation

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\Big(\frac{1}{e}\, g_{\lambda\nu}\dot{x}^{\nu}\Big) \;=\; \frac{1}{2e}\,(\partial_{\lambda} g_{\mu\nu})\,\dot{x}^{\mu}\dot{x}^{\nu}. \tag{38}$$

We now convert (38) into an equivalent form by a sequence of elementary calculus steps.

Step 1. The left-hand side of (38) is the ordinary derivative of a product of two τ-dependent factors:

$$\frac{1}{e(\tau)} \qquad \text{and} \qquad g_{\lambda\nu}(x(\tau))\,\dot{x}^{\nu}(\tau).$$

Apply the ordinary product rule:

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\left(\frac{1}{e}\, g_{\lambda\nu}\dot{x}^{\nu}\right) = \left(\frac{\mathrm{d}}{\mathrm{d}\tau}\frac{1}{e}\right)(g_{\lambda\nu}\dot{x}^{\nu}) + \frac{1}{e}\,\frac{\mathrm{d}}{\mathrm{d}\tau}(g_{\lambda\nu}\dot{x}^{\nu}). \tag{39}$$

Step 2. Differentiate the reciprocal of the einbein by the chain rule:

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\left(\frac{1}{e}\right) = \frac{\mathrm{d}}{\mathrm{d}\tau}(e^{-1}) = -e^{-2}\,\dot{e} = -\frac{\dot{e}}{e^{2}}. \tag{40}$$

Step 3. Substitute the result of Step 2 back into the product rule of Step 1:

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\left(\frac{1}{e}\, g_{\lambda\nu}\dot{x}^{\nu}\right) = \left(-\frac{\dot{e}}{e^{2}}\right)(g_{\lambda\nu}\dot{x}^{\nu}) + \frac{1}{e}\,\frac{\mathrm{d}}{\mathrm{d}\tau}(g_{\lambda\nu}\dot{x}^{\nu}). \tag{41}$$

Step 4. Equation (38) therefore reads

$$-\frac{\dot{e}}{e^{2}}\, g_{\lambda\nu}\dot{x}^{\nu} + \frac{1}{e}\,\frac{\mathrm{d}}{\mathrm{d}\tau}(g_{\lambda\nu}\dot{x}^{\nu}) = \frac{1}{2e}\,(\partial_{\lambda} g_{\mu\nu})\,\dot{x}^{\mu}\dot{x}^{\nu}. \tag{42}$$

Step 5. Multiply every term of the equation by the positive factor e (which never vanishes by definition of the einbein). The first term becomes $-\frac{\dot{e}}{e}\, g_{\lambda\nu}\dot{x}^{\nu}$, the second term becomes $\frac{\mathrm{d}}{\mathrm{d}\tau}(g_{\lambda\nu}\dot{x}^{\nu})$, and the right-hand side becomes $\frac{1}{2}\,(\partial_{\lambda} g_{\mu\nu})\,\dot{x}^{\mu}\dot{x}^{\nu}$.

Step 6. Rearrange the resulting terms to obtain

$$\frac{\mathrm{d}}{\mathrm{d}\tau}(g_{\lambda\nu}\dot{x}^{\nu}) - \frac{\dot{e}}{e}\, g_{\lambda\nu}\dot{x}^{\nu} = \frac{1}{2}\,(\partial_{\lambda} g_{\mu\nu})\,\dot{x}^{\mu}\dot{x}^{\nu}. \tag{43}$$

All six steps are ordinary calculus (product rule and chain rule) together with multiplication by a nowhere-vanishing positive function; no geometric identity is required for this particular transition.

8.4 Expansion of the total derivative and introduction of the Christoffel symbols

Expand the total derivative that appears on the left-hand side of (43). Write the product as

$$g_{\lambda\nu}(x(\tau))\,\dot{x}^{\nu}(\tau) \tag{44}$$

and apply the ordinary product rule of one-variable calculus:

$$\frac{\mathrm{d}}{\mathrm{d}\tau}(g_{\lambda\nu}\dot{x}^{\nu}) = \left(\frac{\mathrm{d}}{\mathrm{d}\tau}\, g_{\lambda\nu}\right)\dot{x}^{\nu} + g_{\lambda\nu}\,\ddot{x}^{\nu}. \tag{45}$$

The second term is immediate. The first term requires care because the metric components are functions of the spacetime coordinates, which themselves depend on τ. In $1+1$ dimensions this dependence may be written explicitly:

$$g_{\lambda\nu} = g_{\lambda\nu}\big(x^0(\tau), x^1(\tau)\big). \tag{46}$$

The ordinary chain rule of multivariable calculus then states

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\, g_{\lambda\nu} = \frac{\partial g_{\lambda\nu}}{\partial x^0}\,\dot{x}^0 + \frac{\partial g_{\lambda\nu}}{\partial x^1}\,\dot{x}^1. \tag{47}$$

In index notation the sum over the two coordinates is abbreviated

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\, g_{\lambda\nu} = (\partial_\rho g_{\lambda\nu})\,\dot{x}^\rho, \tag{48}$$

where $\partial_\rho = \partial/\partial x^\rho$ and the repeated index ρ is summed. Multiplying by the remaining velocity $\dot{x}^\nu$ produces the quadratic term

$$\left(\frac{\mathrm{d}}{\mathrm{d}\tau}\, g_{\lambda\nu}\right)\dot{x}^\nu = (\partial_\rho g_{\lambda\nu})\,\dot{x}^\rho\dot{x}^\nu. \tag{49}$$

Collecting both contributions yields the expansion used in the text:

$$\frac{\mathrm{d}}{\mathrm{d}\tau}(g_{\lambda\nu}\dot{x}^\nu) = \ (\partial_\rho g_{\lambda\nu})\,\dot{x}^\rho\,\dot{x}^\nu + g_{\lambda\nu}\,\ddot{x}^\nu. \tag{50}$$

(The appearance of the factor $\partial_\rho g_{\lambda\nu}$ in front of the metric is therefore nothing but the first factor produced by the multivariable chain rule; no geometric identity is involved at this stage.)

Substitution of this expansion into (43) produces

$$(\partial_\rho g_{\lambda\nu})\dot{x}^\rho\dot{x}^\nu + g_{\lambda\nu}\ddot{x}^\nu - \frac{\dot{e}}{e}\, g_{\lambda\nu}\dot{x}^\nu \ = \ \frac{1}{2}\,(\partial_\lambda g_{\mu\nu})\,\dot{x}^\mu\dot{x}^\nu. \tag{51}$$

We now raise the free (covariant) index λ by a sequence of purely algebraic steps that use only the inverse metric.

Step 1. Move every term of (51) to the left-hand side:

$$g_{\lambda\nu}\ddot{x}^\nu + (\partial_\rho g_{\lambda\nu})\dot{x}^\rho\dot{x}^\nu - \frac{1}{2}\,(\partial_\lambda g_{\mu\nu})\,\dot{x}^\mu\dot{x}^\nu - \frac{\dot{e}}{e}\, g_{\lambda\nu}\dot{x}^\nu = 0. \tag{52}$$

Step 2. Multiply the entire equation by the inverse-metric component $g^{\sigma\lambda}$ and sum over λ:

$$g^{\sigma\lambda}\Big(g_{\lambda\nu}\ddot{x}^\nu + (\partial_\rho g_{\lambda\nu})\dot{x}^\rho\dot{x}^\nu - \frac{1}{2}\,(\partial_\lambda g_{\mu\nu})\,\dot{x}^\mu\dot{x}^\nu - \frac{\dot{e}}{e}\, g_{\lambda\nu}\dot{x}^\nu\Big) = 0. \tag{53}$$

Step 3. Distribute the factor $g^{\sigma\lambda}$ term by term.

- First term (second derivatives):

  $$g^{\sigma\lambda}g_{\lambda\nu}\ddot{x}^\nu = \delta^\sigma{}_\nu\ddot{x}^\nu = \ddot{x}^\sigma \tag{54}$$

  (by the defining property of the inverse metric).

- Last term (einbein derivative):

  $$g^{\sigma\lambda}\Big(-\frac{\dot{e}}{e}\, g_{\lambda\nu}\dot{x}^\nu\Big) = -\frac{\dot{e}}{e}\,(g^{\sigma\lambda}g_{\lambda\nu})\dot{x}^\nu = -\frac{\dot{e}}{e}\,\delta^\sigma{}_\nu\dot{x}^\nu = -\frac{\dot{e}}{e}\,\dot{x}^\sigma. \tag{55}$$

- The two middle terms that involve derivatives of the metric remain

  $$g^{\sigma\lambda}\Big((\partial_\rho g_{\lambda\nu})\dot{x}^\rho\dot{x}^\nu - \frac{1}{2}\,(\partial_\lambda g_{\mu\nu})\,\dot{x}^\mu\dot{x}^\nu\Big). \tag{56}$$

Step 4. Because the velocity bilinears are symmetric, the combination of metric derivatives that multiplies $\dot{x}^\rho \dot{x}^\nu$ may be rewritten

$$\frac{1}{2}\, g^{\sigma\lambda}(2\partial_\rho g_{\lambda\nu} - \partial_\lambda g_{\rho\nu})\dot{x}^\rho \dot{x}^\nu. \tag{57}$$

(The factor $\frac{1}{2}$ and the rearrangement of indices are purely algebraic; they prepare the expression for the later identification with the Christoffel symbol.)

Step 5. Collecting all pieces yields

$$\ddot{x}^\sigma + \frac{1}{2}\, g^{\sigma\lambda}(2\partial_\rho g_{\lambda\nu} - \partial_\lambda g_{\rho\nu})\dot{x}^\rho \dot{x}^\nu - \frac{\dot{e}}{e}\,\dot{x}^\sigma = 0. \tag{58}$$

Every step above is ordinary linear algebra with the inverse metric (the defining relation $g^{\sigma\lambda}g_{\lambda\nu} = \delta^\sigma{}_\nu$) together with the symmetry of the velocity bilinears; no geometric identity beyond the existence of the inverse metric is required.

The combination of metric derivatives that multiplies $\dot{x}^\rho \dot{x}^\nu$ is precisely the Christoffel symbol of the Levi-Civita connection. In $1+1$ dimensions the independent components are

$$\Gamma^0{}_{00} = \frac{1}{2}g^{00}(\partial_0 g_{00}) + \frac{1}{2}g^{01}(2\partial_0 g_{10} - \partial_1 g_{00}), \tag{59}$$

$$\begin{aligned}\Gamma^0{}_{01} = \Gamma^0{}_{10} &= \frac{1}{2}g^{00}(\partial_1 g_{00}) + \frac{1}{2}g^{01}(\partial_0 g_{11} + \partial_1 g_{01} - \partial_1 g_{01}) \\ &= \frac{1}{2}g^{00}\partial_1 g_{00} + \frac{1}{2}g^{01}\partial_0 g_{11},\end{aligned} \tag{60}$$

$$\Gamma^0{}_{11} = \frac{1}{2}g^{00}(2\partial_1 g_{01} - \partial_0 g_{11}) + \frac{1}{2}g^{01}(\partial_1 g_{11}), \tag{61}$$

$$\Gamma^1{}_{00} = \frac{1}{2}g^{10}(\partial_0 g_{00}) + \frac{1}{2}g^{11}(2\partial_0 g_{10} - \partial_1 g_{00}), \tag{62}$$

$$\Gamma^1{}_{01} = \Gamma^1{}_{10} = \frac{1}{2}g^{10}(\partial_1 g_{00}) + \frac{1}{2}g^{11}(\partial_0 g_{11}), \tag{63}$$

$$\Gamma^1{}_{11} = \frac{1}{2}g^{10}(2\partial_1 g_{01} - \partial_0 g_{11}) + \frac{1}{2}g^{11}(\partial_1 g_{11}). \tag{64}$$

(The general formula used above is the standard definition

$$\Gamma^\sigma{}_{\rho\nu} \;=\; \frac{1}{2}g^{\sigma\lambda}(\partial_\rho g_{\lambda\nu} + \partial_\nu g_{\lambda\rho} - \partial_\lambda g_{\rho\nu}). \tag{65}$$

Because the lower indices are symmetric, $\Gamma^\sigma{}_{\rho\nu} = \Gamma^\sigma{}_{\nu\rho}$, only the six components listed need be written; the remaining two are identical by symmetry.)

With these symbols the Euler–Lagrange equation takes the compact form

$$\ddot{x}^\sigma + \Gamma^\sigma{}_{\rho\nu}\,\dot{x}^\rho \dot{x}^\nu - \frac{\dot{e}}{e}\,\dot{x}^\sigma \;=\; 0, \tag{66}$$

or, equivalently,

$$\frac{\mathrm{D}}{\mathrm{d}\tau}\Big(\frac{\dot{x}^\sigma}{e}\Big) \;=\; 0. \tag{67}$$

Here the symbol $\dfrac{\mathrm{D}}{\mathrm{d}\tau}$ denotes the *covariant* (or absolute) derivative along the world-line. By definition,

$$\frac{\mathrm{D}V^\sigma}{\mathrm{d}\tau} \;:=\; \frac{\mathrm{d}V^\sigma}{\mathrm{d}\tau} + \Gamma^\sigma{}_{\rho\nu}\,\dot{x}^\rho V^\nu \tag{68}$$

for any vector field V^σ defined along the curve. Only the numerator carries a capital D by long-standing mathematical convention; the differential of the parameter remains the ordinary $\mathrm{d}\tau$. (Some older texts write $\frac{\mathrm{D}}{\mathrm{D}\tau}$, but the form used here is the modern standard; see e.g. Wald, *General Relativity*, 1984, p. 34, or Carroll, *Spacetime and Geometry*, 2004, §3.3.)

Equation (66) is the geodesic equation written with respect to a *non-affine* parameter.[2] The term proportional to $\dot{e}/e$ does **not** disappear automatically; it is the precise measure of the failure of τ to be an affine parameter. We now explain, step by step, why the term is present and how it can be removed by a legitimate choice of the free parameter.

Step 1 – Residual reparametrisation freedom

The world-line parameter τ that appears in the action is still completely arbitrary. The original einbein action is invariant under the infinite-dimensional group of orientation-preserving reparametrisations[3]

$$\tau \;\mapsto\; f(\tau), \qquad e(\tau) \;\mapsto\; \frac{e(\tau)}{\dot{f}(\tau)} \quad (\dot{f} > 0). \tag{69}$$

This residual freedom has not yet been fixed. Consequently the equation of motion still contains a term that vanishes if and only if the chosen parameter happens to be affine.

Step 2 – The algebraic (einbein) constraint

Independently of the choice of parameter, the Euler–Lagrange equation obtained by varying the action with respect to the einbein[4] is the purely algebraic relation

$$g_{\mu\nu}\dot{x}^{\mu}\dot{x}^{\nu} \;=\; -e^2(mc_0)^2. \tag{70}$$

It is called an *algebraic constraint*[5] (or simply the *einbein constraint*) because it involves no derivative of e.

Step 3 – Imposing a convenient gauge condition

Because of the residual reparametrisation freedom one is allowed to impose one additional condition on the variables. The most convenient choice is the gauge condition[6]

$$e(\tau) \;=\; \text{constant} \qquad \Rightarrow \qquad \dot{e} = 0. \tag{71}$$

(This gauge is always attainable: given any positive function $e(\tau)$ one can solve the ordinary differential equation that defines the new parameter f so that the transformed einbein is constant.)

Step 4 – Disappearance of the non-affine term

Once the gauge condition of Step 3 is imposed, the coefficient of the unwanted term vanishes identically:

$$\frac{\dot{e}}{e}\,\dot{x}^{\sigma} \;=\; 0. \tag{72}$$

Equation (66) therefore collapses to the **canonical affine geodesic equation**

$$\frac{\mathrm{D}\dot{x}^{\sigma}}{\mathrm{d}\tau} = \ddot{x}^{\sigma} + \Gamma^{\sigma}{}_{\rho\nu}\,\dot{x}^{\rho}\dot{x}^{\nu} = 0. \tag{73}$$

[2]A parameter τ along a curve is called *affine* if the geodesic equation takes the simple form $\frac{\mathrm{D}\dot{x}^{\sigma}}{\mathrm{d}\tau} = 0$. Any other parameter produces an extra term proportional to $\dot{x}^{\sigma}$; such a parameter is called *non-affine*.

[3]"Reparametrisation freedom" (also called residual gauge freedom) means that if $(x(\tau), e(\tau))$ is a solution of the equations of motion, then so is the transformed pair $\big(x(f(\tau)),\, e(f(\tau))/\dot{f}(\tau)\big)$ for any strictly increasing function f. The physical world-line is the same; only the numerical label τ changes.

[4]"Variation of the einbein" means the ordinary calculus-of-variations procedure: one changes $e(\tau)$ by an arbitrary infinitesimal function $\delta e(\tau)$ while keeping the world-line $x^{\mu}(\tau)$ fixed, and one requires that the first-order change in the action vanishes. Because the Lagrangian does not depend on $\dot{e}$, the resulting equation contains no derivatives of e.

[5]An equation is called algebraic (as opposed to differential) when it does not involve derivatives of the variable that is being solved for. Here the variable is e itself; the equation determines the numerical value of e at each τ once x and $\dot{x}$ are known. It is a *constraint* because it restricts the admissible pairs $(x, \dot{x})$.

[6]In the language of constrained systems a *gauge condition* (or gauge-fixing condition) is an extra equation that one is free to impose in order to remove the redundancy associated with a local symmetry. It is not an equation of motion; it is a choice of representative inside each equivalence class of reparametrisations.

Step 5 – Interpretation after the gauge choice

With e now a constant, the algebraic constraint (70) becomes a simple normalisation of the tangent vector:

$$g_{\mu\nu}\dot{x}^{\mu}\dot{x}^{\nu} \;=\; -\text{const} < 0. \tag{74}$$

This shows that the parameter τ is proportional to the particle's proper time (the constant of proportionality being $1/(e\,mc_0)$).

Thus the single gauge choice $e = \text{const}$ has achieved two things at once:

- it has eliminated the non-affine term, converting the equation of motion into the standard affine geodesic equation;
- it has identified the world-line parameter with (a multiple of) proper time.

Summary of the logical chain

Stage	Condition imposed	Resulting equation
before gauge fixing	none (arbitrary τ)	non-affine form (66)
after gauge fixing	$e = \text{const}$ (so $\dot{e} = 0$)	affine geodesic equation + mass-shell constraint

The term $\dot{e}/e\,\dot{x}^{\sigma}$ is therefore not a defect of the derivation; it is the signal that the residual reparametrisation freedom has not yet been used. Once that freedom is used to impose the convenient gauge $e = \text{const}$, the term vanishes and the standard geodesic equation appears.

The same chain of reasoning applies verbatim to the massless theory: the only change is that the algebraic constraint becomes the null condition $g_{\mu\nu}\dot{x}^{\mu}\dot{x}^{\nu} = 0$ instead of the massive mass-shell condition.

Thus the Euler–Lagrange equations of the einbein action, after the einbein constraint is taken into account and a convenient gauge is chosen, are exactly the geodesic equations of the background metric.

9 The Massless Case: Euler–Lagrange Equations and Null Geodesics

We now set the rest mass to zero, $m = 0$. The Lagrangian simplifies to

$$L \;=\; \frac{1}{2e}\,g_{\mu\nu}(x)\,\dot{x}^{\mu}\dot{x}^{\nu}. \tag{75}$$

The action is therefore

$$S[x,e] \;=\; \int \frac{1}{2e}\,g_{\mu\nu}(x)\,\dot{x}^{\mu}\dot{x}^{\nu}\,\mathrm{d}\tau. \tag{76}$$

9.1 Euler–Lagrange equation for the einbein (massless)

Because L still does not depend on $\dot{e}$, the equation $\partial L/\partial e = 0$ reduces to

$$-\frac{1}{2e^2}\,g_{\mu\nu}\dot{x}^{\mu}\dot{x}^{\nu} \;=\; 0, \tag{77}$$

which immediately implies the null constraint

$$g_{\mu\nu}\dot{x}^{\mu}\dot{x}^{\nu} \;=\; 0. \tag{78}$$

9.2 Euler–Lagrange equations for the coordinates (massless)

The partial derivatives with respect to velocity and coordinate are identical in form to those of the massive case (the mass term is absent, but it never contributed to these derivatives). Consequently one obtains again

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\Big(\frac{1}{e}\,g_{\lambda\nu}\dot{x}^{\nu}\Big) \;=\; \frac{1}{2e}\,(\partial_{\lambda}g_{\mu\nu})\,\dot{x}^{\mu}\dot{x}^{\nu}. \tag{79}$$

Exactly the same algebraic steps that led from (38) to (66) now produce

$$\ddot{x}^{\sigma} + \Gamma^{\sigma}{}_{\rho\nu}\,\dot{x}^{\rho}\dot{x}^{\nu} - \frac{\dot{e}}{e}\,\dot{x}^{\sigma} \;=\; 0, \tag{80}$$

where the Christoffel symbols are still given by the explicit expressions (59)–(64).

Removal of the non-affine term in the massless case

The reasoning developed for the massive particle applies verbatim and can be reused without change. The residual reparametrisation invariance of the action is identical:

$$\tau \;\mapsto\; f(\tau), \qquad e(\tau) \;\mapsto\; \frac{e(\tau)}{\dot{f}(\tau)} \quad (\dot{f} > 0). \tag{81}$$

Consequently one may always choose a new parameter in which the einbein is constant,

$$e(\tau) \;=\; \text{const} \qquad \Rightarrow \qquad \dot{e} = 0. \tag{82}$$

In that gauge the non-affine term vanishes identically and equation (80) collapses to the **affine null-geodesic equation**

$$\frac{\mathrm{D}\dot{x}^{\sigma}}{\mathrm{d}\tau} \;=\; 0, \qquad g_{\mu\nu}\dot{x}^{\mu}\dot{x}^{\nu} \;=\; 0. \tag{83}$$

(The only difference with respect to the massive theory is the algebraic constraint enforced by the einbein equation of motion: it is now the null condition rather than the massive mass-shell condition. The residual gauge freedom that removes the $\dot{e}$ term is exactly the same.)

Thus, in the massless theory as well, the Euler–Lagrange equations of the einbein action are precisely the geodesic equations of the background metric (now restricted to the null cone).

The massless limit is therefore completely regular: the action (76) is well-defined, the constraint (78) is enforced dynamically by the variation of the einbein, and the resulting trajectories are exactly the null geodesics.

10 Brief Hamiltonian Perspective

For completeness we record the Hamiltonian formulation that follows from the einbein Lagrangian. The canonical momenta conjugate to the coordinates are

$$p_{\lambda} \;=\; \frac{\partial L}{\partial \dot{x}^{\lambda}} \;=\; \frac{1}{e}\,g_{\lambda\nu}\dot{x}^{\nu}. \tag{84}$$

Inverting for the velocities,

$$\dot{x}^{\nu} \;=\; e\,g^{\nu\lambda}p_{\lambda}. \tag{85}$$

The Hamiltonian (after a Legendre transform) is pure constraint:

$$H \;=\; \frac{e}{2}\big(g^{\mu\nu}p_{\mu}p_{\nu} + (mc_0)^2\big). \tag{86}$$

The einbein itself acts as a Lagrange multiplier enforcing the mass-shell condition

$$g^{\mu\nu}p_{\mu}p_{\nu} + (mc_0)^2 \;=\; 0. \tag{87}$$

In the massless case this reduces to the light-like condition $p^2 = 0$. This constrained Hamiltonian system is the starting point for canonical quantization of the relativistic particle (see, e.g., the references listed below).

11 Resolution of the Equations of Motion and the Resulting Trajectories

After the residual reparametrisation freedom has been used to set $e = \text{const}$, the dynamical content of the theory is completely determined. We summarise the final system of equations and the nature of its solutions for both the massive and the massless particle.

11.1 Massive particle ($m > 0$)

In the gauge $e = \text{const}$ the Euler–Lagrange equations reduce to the pair

$$\frac{\mathrm{D}\dot{x}^\sigma}{\mathrm{d}\tau} = \ddot{x}^\sigma + \Gamma^\sigma{}_{\rho\nu}\,\dot{x}^\rho\dot{x}^\nu = 0, \tag{88}$$

$$g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = -e^2(mc_0)^2 = \text{const} < 0. \tag{89}$$

Equation (88) is the affine geodesic equation; equation (89) is the mass-shell (or normalisation) constraint.

General-metric solution. Any solution of the system (88)–(89) is, by definition, a future-directed timelike geodesic of the background metric $g_{\mu\nu}$, parametrised proportionally to proper time.

The initial-value problem is well-posed. Given data

$$x^\mu(0) = x_0^\mu, \qquad \dot{x}^\mu(0) = u_0^\mu$$

that satisfy the mass-shell condition at $\tau = 0$, the Picard–Lindelöf theorem for ordinary differential equations guarantees a unique local solution $x^\mu(\tau)$. The solution extends to a maximal interval of existence and is global if the spacetime is geodesically complete.

In $1+1$ dimensions a further reduction is always available. The algebraic constraint (89) can be solved for one of the two velocity components in terms of the other and of the metric coefficients. Substituting the result into either component of the geodesic equation yields a single first-order ordinary differential equation for the remaining velocity (or, equivalently, a first-order equation for the orbit $x^1 = x^1(x^0)$). Once a concrete metric $g_{\mu\nu}(x)$ is given, this reduced equation can in principle be integrated by quadrature.

If, in addition, the metric admits a Killing vector field ξ^μ, the quantity

$$E = g_{\mu\nu}\xi^\mu\dot{x}^\nu$$

is a constant of motion. The first integral $E = \text{const}$ supplies a further algebraic relation that reduces the problem to a single quadrature.

When no Killing vector is present and the metric is completely generic, the geodesic equation (or the reduced first-order system obtained from the mass-shell constraint) is itself the most explicit form of the trajectory. The solution may still be written formally by means of the exponential map of the Levi-Civita connection:

$$x(\tau) = \exp_{x_0}(\tau\, u_0), \tag{90}$$

where $\exp_{x_0} : T_{x_0}\mathcal{M} \to \mathcal{M}$ is the exponential map based at the initial point x_0.

Flat-space illustration. When $g_{\mu\nu} = \eta_{\mu\nu}$ (Minkowski metric) the Christoffel symbols vanish and the equations become

$$\ddot{x}^\sigma = 0, \qquad \eta_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = -e^2(mc_0)^2.$$

The general solution is the straight line

$$x^\sigma(\tau) = x_0^\sigma + u_0^\sigma\tau,$$

with the constant four-velocity satisfying $\eta_{\mu\nu}u_0^\mu u_0^\nu = -e^2(mc_0)^2$. This is the special case of the exponential map in which $\exp_{x_0}$ is ordinary vector addition.

11.2 Massless particle ($m = 0$)

In the same gauge $e = \text{const}$ the equations reduce to

$$\frac{\mathrm{D}\dot{x}^\sigma}{\mathrm{d}\tau} = \ddot{x}^\sigma + \Gamma^\sigma{}_{\rho\nu}\,\dot{x}^\rho \dot{x}^\nu = 0, \tag{91}$$

$$g_{\mu\nu}\dot{x}^\mu \dot{x}^\nu = 0. \tag{92}$$

Equation (91) is again the affine geodesic equation; equation (92) is the null constraint.

General-metric solution. Any solution of the system (91)–(92) is a null (light-like) geodesic of the background metric, parametrised affinely.

The initial-value problem is again well-posed. Given data

$$x^\mu(0) = x_0^\mu, \qquad \dot{x}^\mu(0) = k_0^\mu$$

that satisfy the null condition at $\tau = 0$, there exists a unique local solution $x^\mu(\tau)$, which extends to a maximal interval and is global if the spacetime is null-geodesically complete.

In $1+1$ dimensions the null constraint can be solved for one velocity component in terms of the other and of the metric coefficients. Substitution into the geodesic equation yields a single first-order ODE that determines the null orbit. Once a concrete metric is given, the equation can be integrated by quadrature.

If a Killing vector ξ^μ is present, the conserved quantity $g_{\mu\nu}\xi^\mu \dot{x}^\nu$ supplies an additional first integral and reduces the problem further. In the absence of symmetries the formal solution is again given by the exponential map

$$x(\tau) = \exp_{x_0}(\tau\, k_0) \tag{93}$$

restricted to the null cone of $T_{x_0}\mathcal{M}$.

Flat-space illustration. In Minkowski space the equations become

$$\ddot{x}^\sigma = 0, \qquad \eta_{\mu\nu}\dot{x}^\mu \dot{x}^\nu = 0.$$

The general solution is the straight null line

$$x^\sigma(\tau) = x_0^\sigma + k_0^\sigma \tau,$$

with the constant null vector satisfying $\eta_{\mu\nu}k_0^\mu k_0^\nu = 0$. This recovers the ordinary exponential map of a vector space restricted to the light cone.

11.3 Summary of the trajectories

Case	Final equations	Geometric character of the trajectory
Massive ($m > 0$)	affine geodesic eq. + mass-shell	future-directed timelike geodesic
Massless ($m = 0$)	affine geodesic eq. + null condition	null (light-like) geodesic

In both cases the einbein formulation has achieved its purpose: the equations of motion are well-defined, the massless limit is continuous, and the solutions are precisely the geodesics of the appropriate causal type.

12 Summary

We have shown, by completely explicit algebraic manipulations in $1 + 1$ dimensions, that the introduction of a positive einbein $e(\tau)$ converts the non-quadratic square-root action of a relativistic particle into a quadratic action that

- is classically equivalent to the original formulation for any positive rest mass m,

- admits a smooth and well-defined limit $m \to 0$,
- yields the correct null-geodesic dynamics in that limit.

All Euler–Lagrange equations were derived term by term, the algebraic solution for the einbein was substituted back into the action, and the recovery of the square-root Lagrangian was verified by direct calculation. The same construction extends without change of principle to higher-dimensional spacetimes and forms the conceptual foundation for the Polyakov formulation of the bosonic string.

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End of tutorial notes.